

## EXPERIMENT: 10

### 5G NETWORK SLICING ENVIRONMENT USING MATLAB

**Objective – 1: Analyze the Number of Network Slices**

MATLAB Code:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

grid on

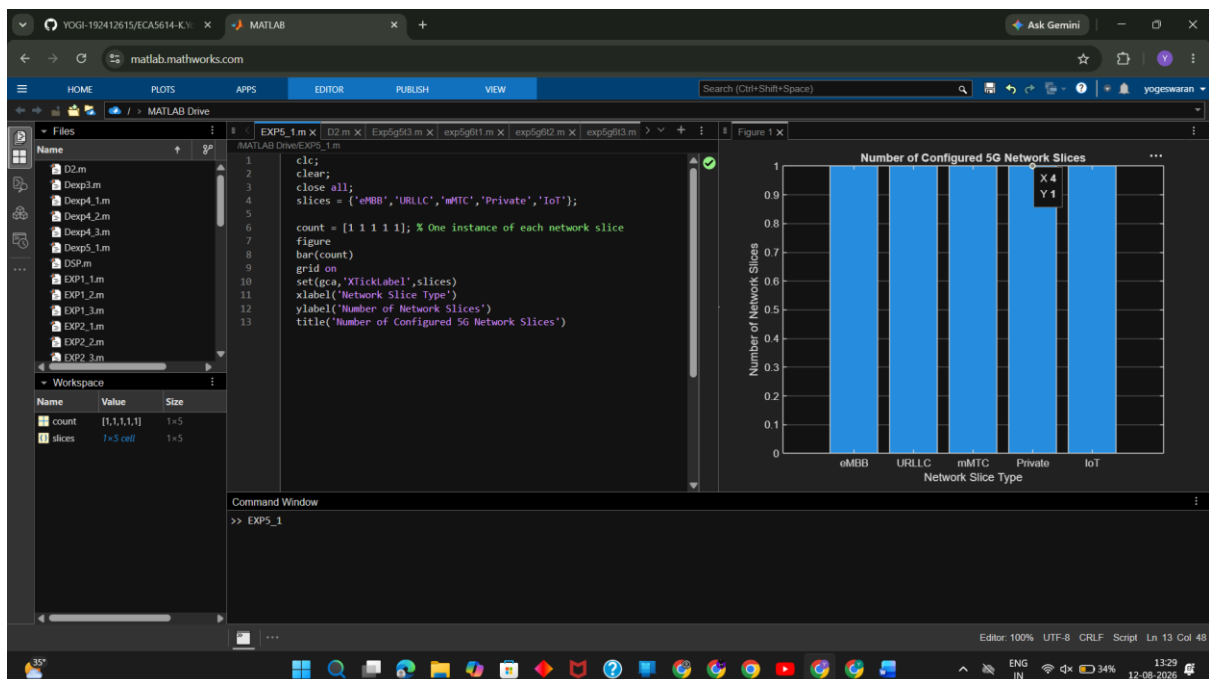
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```

OUTPUT



## Objective – 2: Compare Slice Utilization (%)

Matlab Code :

```
clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

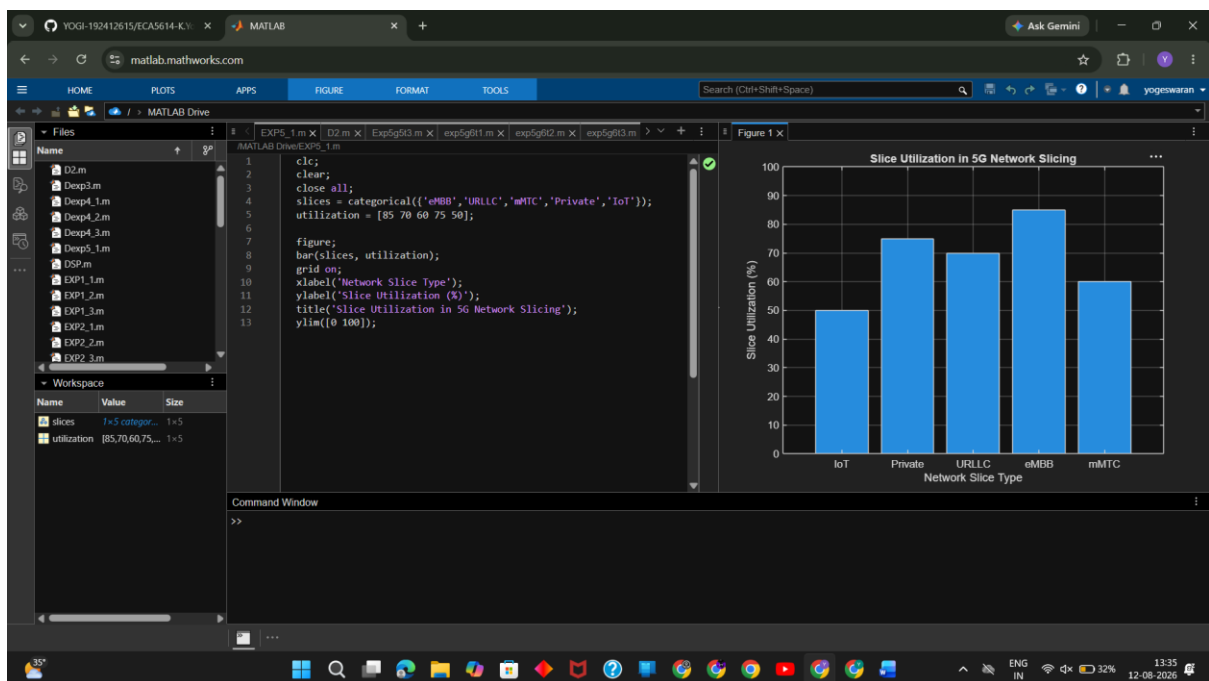
xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);
```

## OUTPUT



## Objective – 3: Analyze End-to-End Latency (ms)

MATLAB Code:

```
clc;

clear;

close all;
```

```

slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;

xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');

```

## OUTPUT

